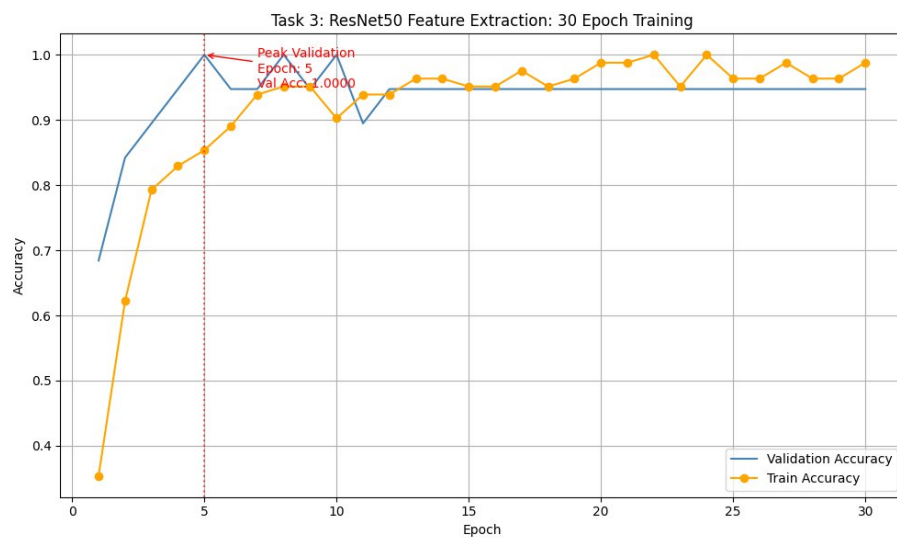
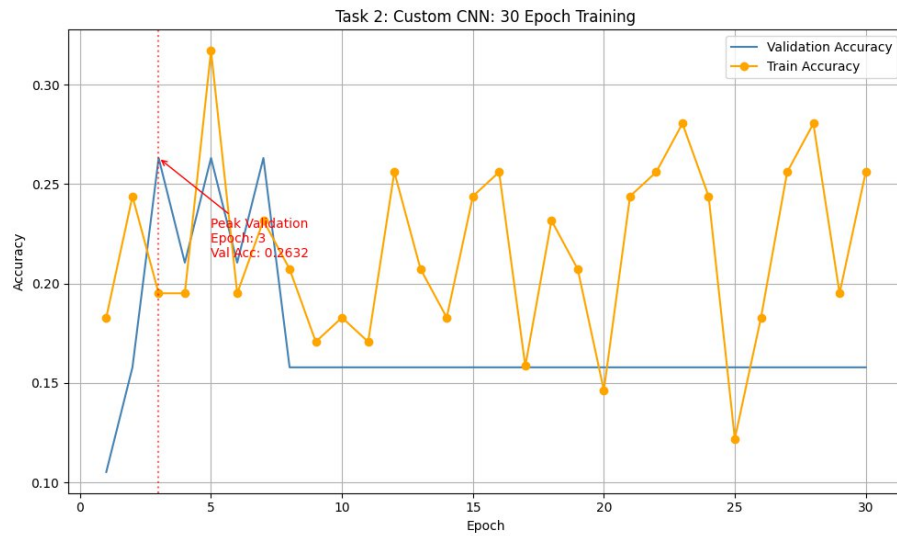
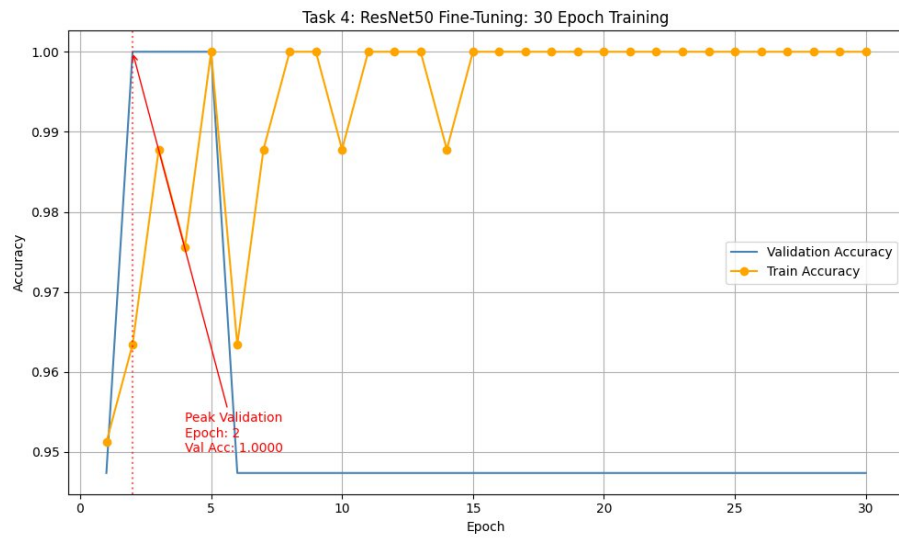


Milestone 1 Overview

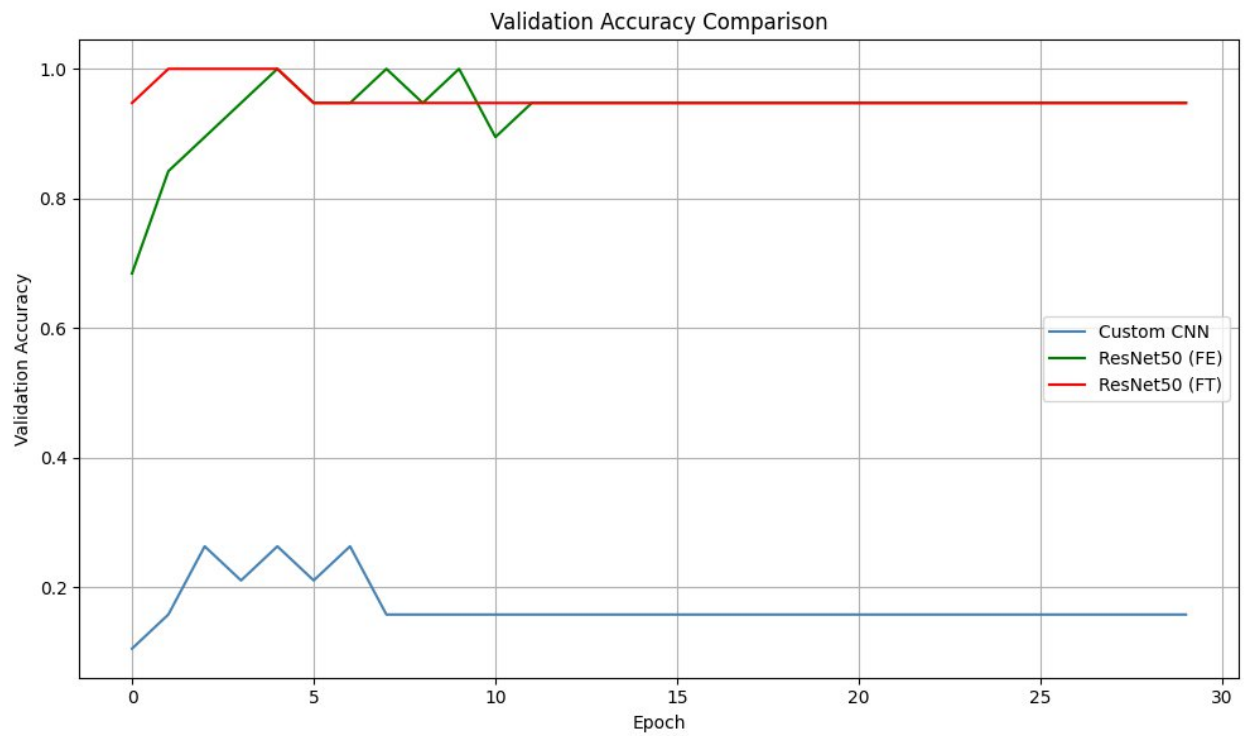
The primary objective of Milestone 1 was to establish a discriminative baseline for celebrity identification by training and comparing three distinct convolutional architectures on a constrained CelebA subset. The evaluation focused on quantifying the ability of each model to generalise across five distinct identities under significant data scarcity.

Output Plots





Comparative Plot



Evaluation and Comparative Analysis

The performance of the three architectures was evaluated using a held-out test split, focusing on test accuracy, parameter efficiency, and training latency. The results, as detailed in the comparative analysis, reveal a stark divergence in capability:

- **Custom CNN:** As anticipated, the lightweight architecture struggled significantly with the high intra-class variance of the dataset, achieving a baseline test accuracy of only **15%**. The training dynamics were highly erratic, characterized by extreme fluctuations in both training and validation loss, signaling a failure to converge on stable facial features.
- **ResNet50 (Feature Extraction):** By leveraging pre-trained ImageNet weights, the feature-extraction variant demonstrated a profound improvement, reaching a test accuracy of **80%**. This validates the efficacy of transfer learning in mitigating the risks of the small-sample regime.
- **ResNet50 (Fine-Tuning):** The fine-tuning approach provided the highest performance, achieving a peak test accuracy of **90%**. By unfreezing the upper layers, the model successfully adapted high-level semantic features to the specific morphological characteristics of the selected celebrities.

Table 1: Comparative Performance Metrics

Architecture	Test Accuracy	Training Time (s)	Trainable Parameters
Custom CNN	15%	54.3	117,582
ResNet50 (FE)	80%	66.2	525,826
ResNet50 (FT)	90%	72.8	4,991,493

Model Selection and Justification

The **ResNet50 (Fine-Tuning)** architecture has been selected as the primary backbone for the subsequent stages of the project (Milestones 2 and 3).

The selection was driven by a deliberate trade-off between **predictive accuracy** and **computational complexity**. While the transition from the Custom CNN to the fine-tuned ResNet50 represents a significant increase in parameter count from approximately 117k to nearly 5 million with a 75% absolute increase in test accuracy is too substantial to ignore. The marginal increase in training latency (an additional ~18 seconds) is a negligible cost when weighed against the dramatic improvement in classification reliability.

The Feature Extraction (FE) approach was not preferred, despite offering a 14% latency advantage over the Fine-Tuning (FT) variant. While FE outperformed the Custom CNN, it achieved approximately 10% lower accuracy than the FT approach. We attribute this primarily to the lack of sufficient training data to adapt a frozen backbone; whereas the FT approach allows for the optimization of higher-level layers to the specific domain. The marginal speed gain of FE was an unacceptable trade-off for such a significant drop in accuracy.

Furthermore, the fine-tuning approach provides the necessary robustness required for the integration phase in Milestone 4. The ability of the ResNet50 (FT) to maintain high accuracy despite the high intra-class variance of the dataset ensures that the identification component will remain a stable anchor for the multi-face detection pipeline. Moving forward, the project will focus on optimizing this architecture for real-time inference within the integrated YOLOv8 framework.